AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 4 · STUDENT LESSON

Configuration and secrets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Classify the value. Validate what is required. Choose what is emitted.

Separate public configuration from mock server-only inputs and diagnose missing settings without dumping privileged values.

Keep one habit: Input/output contract → startup flow → debug mistake → setting-change picture → probe table → mock audit → independent browser error.

MAIN LESSON · SECTION 1

Load the settings you need without publishing privileged values

> Separate reviewed public configuration from server-only credentials.

Learning goals

I can explain the **mechanism** and terminology of **configuration** and **secret**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve validation and service boundaries

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Typed contracts and schemas. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

Explain **configuration** and **secret**s using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **configuration** and **secret**s
</details>

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Decide which values are public before designing the log

> A setting and an access credential need different handling.

Start with the card-and-key analogy

Imagine a classroom machine with an instruction card and a locked access key. The card can say which lesson-data version to use and how long an answer should be. The key grants privileged access and belongs on a protected path. If the data-version setting is missing, the machine can name the missing setting without displaying every value it can access.

In this lesson both the settings and the private values are invented placeholders.

Define configuration and secrets

Configuration supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

Complete local startup contract

Input or record	Declared role and rule
DATA_VERSION	Reviewed non-secret configuration: nonblank string
MAX_WORDS	Reviewed non-secret configuration: exact positive integer, bool excluded

Input or record	Declared role and rule
SIGNING_SECRET	Server-only mock input: nonblank placeholder; never copy its value into emitted records
Missing/invalid required field	Block startup and report the key name, not its value
Successful public record	Only status, data_version and maximum_words
Internal runtime state	May carry the placeholder separately; never print or serialize that state in the lab
Actual operations	No real environment, credential store, signing, model answer or deployment is used

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Report the missing setting rather than dumping every input

> Fail clearly before startup proceeds with an invented credential.

Original data-version and server-only-secret mechanism

The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

Complete startup and output flow

Check required values, then build a separate public record

STAGE	LOCAL CLASSROOM BEHAVIOUR
1. Read declared inputs	Take DATA_VERSION/MAX_WORDS from a mock config dict; placeholder from a separate server-only dict
2. Validate required values	Missing or invalid keys → startup_blocked with names only; no default credential
3. Hold internal state	Keep the placeholder on the internal path, separate from displayed status
4. Build an allowlisted public record	Ready status + reviewed data version + configured word limit; never dump all inputs
5. Inspect emitted records	Check that the tested mock private values do not appear in any serialized public/error record

This is a local handling exercise. It demonstrates chosen output fields, not a production secret-management system.

State the declared contract before tracing

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Configuration and secrets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A startup script prints every environment variable while debugging a missing data version.	The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.	It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change a public setting without changing the credential

> Predict the configured value and the permitted output fields.

Current task and response

- A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed word-limit picture

- Change: Change maximum_words to 60.

▼ **Check after predicting**

Prediction: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Keep diagnostics useful and private values off the display path

> A successful load and a leak-free complete system are different claims.

Main debug-output mistake and bounded repair

A startup script prints every environment variable while debugging a missing data version.

It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Keep the reasoning’s scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured settings and placeholder

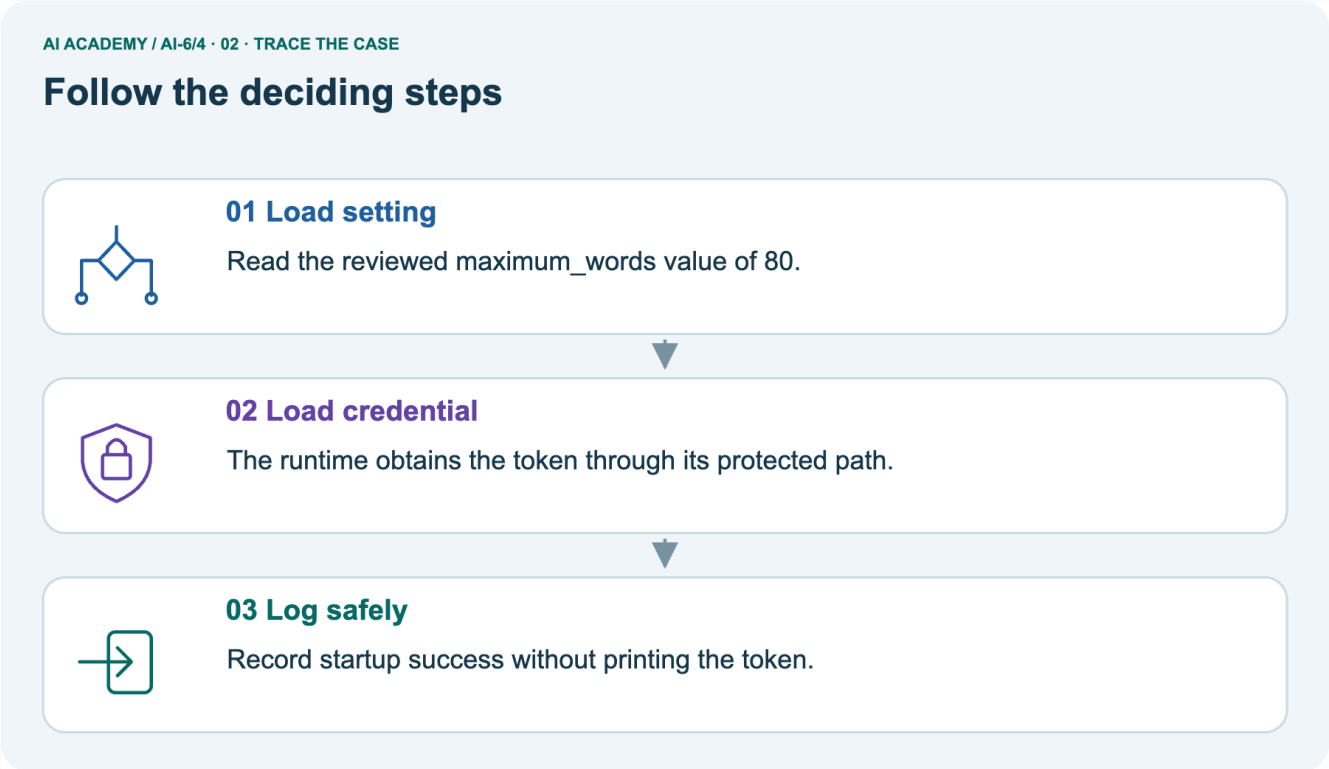
Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

maximum_words=80 is public **configuration**. A separate protected store holds a placeholder credential.

The main failure is a debug script that prints every environment value while diagnosing a missing DATA_VERSION. Replace that output with a missing-key-name record. The picture separately sets maximum_words to 80 with a placeholder credential on a protected path. Changing the configured limit to 60 changes the reviewed setting; it does not require changing the credential.

Our local code records that setting only: no answer is generated, so enforcement of an actual response length has not been measured.

Trace public setting, private path and startup record



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Load setting	Read the reviewed maximum_words value of 80.
2. Load credential	The runtime obtains the token through its protected path.
3. Log safely	Record startup success without printing the token.

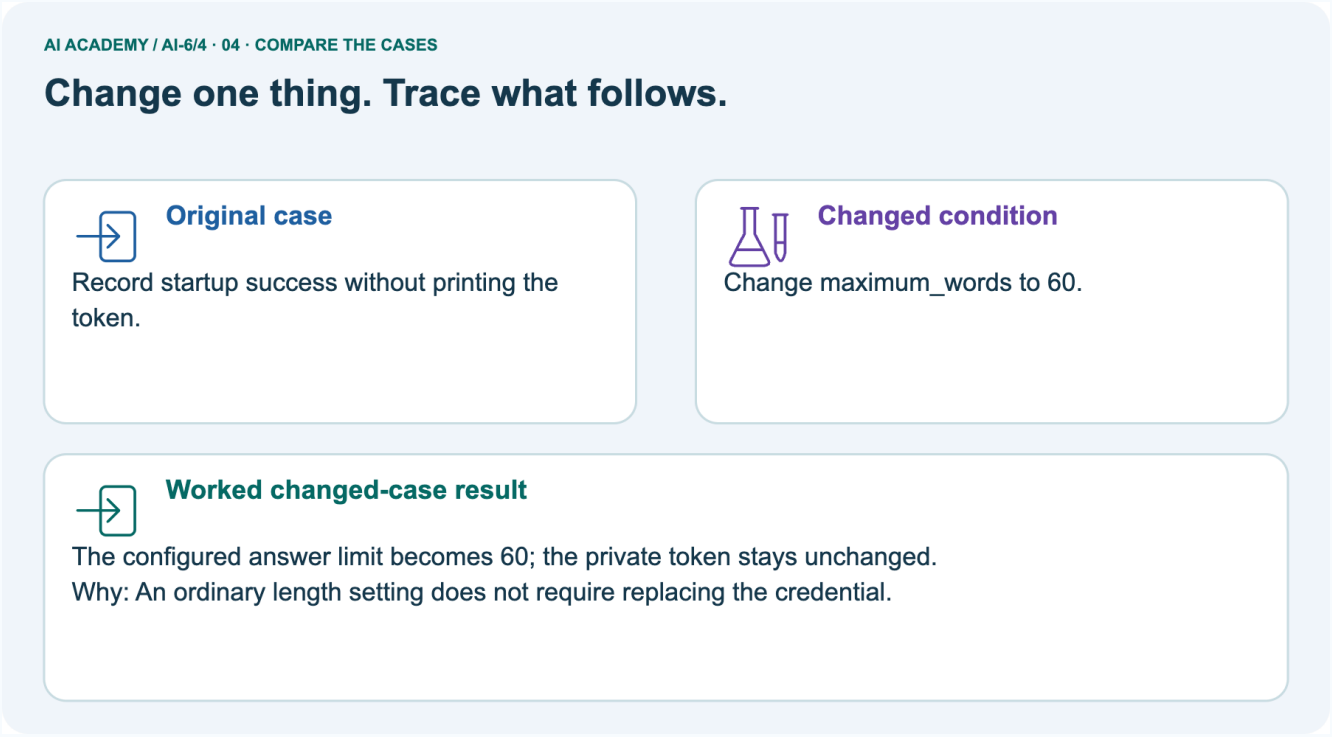
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Configuration controls behaviour. A credential grants access and needs different storage and handling.

Predict changing 80 to 60

- Change one thing: Change maximum_words to 60. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what stays private and remaining leak paths

- Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

- Read the labelled picture: Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

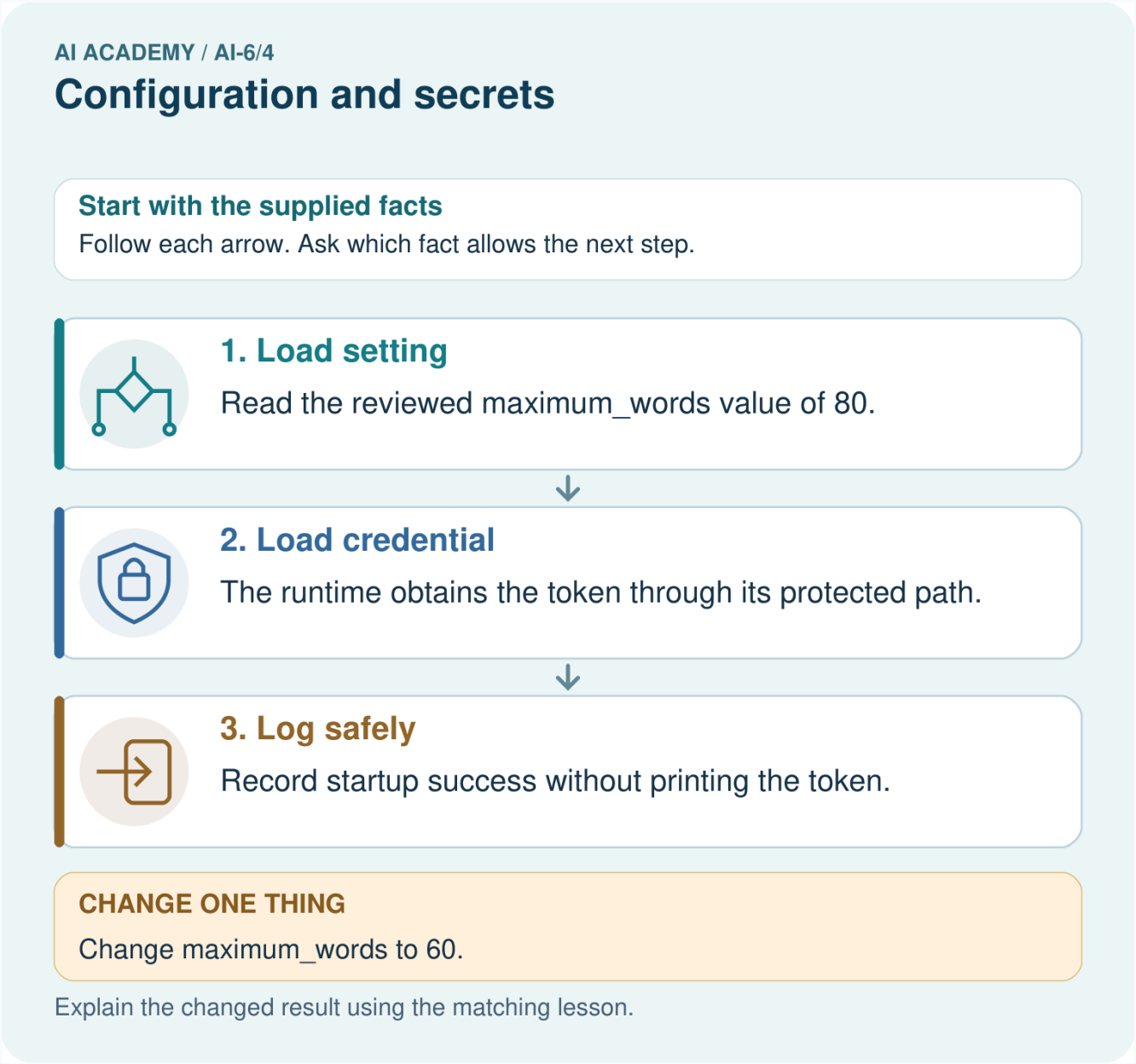
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Configuration and secrets

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Test missing settings and inspect emitted records

> A protected input path is only one part of correct handling.

Core classroom-mock qualification

A classroom mock is not a production credential-management design; no real **secret**s are needed here.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete startup, type and disclosure probes

Complete mock probe	Expected record or scope
DATA_VERSION classroom-v1; MAX_WORDS 80; placeholder present	ready; public version and limit only
DATA_VERSION missing	startup_blocked; missing_or_invalid [DATA_VERSION], without input values
Server-only placeholder absent	startup_blocked; missing_or_invalid [SIGNING_SECRET], without a default
MAX_WORDS True, zero or a quoted number	startup_blocked under the exact positive-integer rule
Limit changes 80 → 60	Public configured limit changes; placeholder unchanged
Unrelated mock private value exists	Never iterate/dump it into the public/error record

Complete mock probe	Expected record or scope
Private store exists	Does not by itself prevent leaks in other logs, screenshots or code paths
Real credentials or deployment	Outside this mock; no real secrets or external operations needed

Guided explanation and repair

- A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise naming the needed setting and safe diagnostic

> Keep key names distinct from their privileged values.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 record

Use workbook W2 for the connected example "**Configuration** and **secret**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual mock-versus-production check

- Use the worked case for **Configuration** and **secret**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured public/private distinction

- Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

> **My explanation using a pictured step and the given facts**
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Reserve the independent browser-error case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete mock startup and output audit

> Use only the supplied fictional placeholders and record exactly what the checks cover.

Current debug-output investigation

A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

Complete runnable mock startup

```
"""Mock dictionaries only: never read real environment variables or credentials."""
import json

def startup(config,server_only):
    invalid=[]
    version=config.get('DATA_VERSION')
    maximum=config.get('MAX_WORDS')
    secret=server_only.get('SIGNING_SECRET')
    if type(version) is not str or not version.strip():invalid.append('DATA_VERSION')
    if type(maximum) is not int or maximum<=0:invalid.append('MAX_WORDS')
    if type(secret) is not str or not secret.strip():invalid.append('SIGNING_SECRET')
    if invalid:
        return None,{'status':'startup_blocked','missing_or_invalid':invalid}
    internal={'data_version':version,'maximum_words':maximum,'signing_secret':secret}
    # Construct a new allowlisted public record, never serialize internal state.
    public={'status':'ready','data_version':version,'maximum_words':maximum}
    return internal,public

config={'DATA_VERSION':'classroom-v1','MAX_WORDS':80}
mock_private={'SIGNING_SECRET':'fictional-placeholder-not-a-real-credential',
              'UNRELATED_SECRET':'unrelated-fictional-sentinel'}
internal,public=startup(config,mock_private)
_,missing=startup({'MAX_WORDS':80},mock_private)
_,absent_secret=startup(config,{})
_,changed=startup({'DATA_VERSION':'classroom-v1','MAX_WORDS':60},mock_private)
assert public=={'status':'ready','data_version':'classroom-v1','maximum_words':80}
assert missing=={'status':'startup_blocked','missing_or_invalid':['DATA_VERSION']}
assert absent_secret=={'status':'startup_blocked','missing_or_invalid':['SIGNING_SECRET']}
assert changed['maximum_words']==60 and
```

```
internal['signing_secret']==mock_private['SIGNING_SECRET']
serialized=json.dumps([public,missing,absent_secret,changed])
assert all(value not in serialized for value in mock_private.values())
assert set(public)=={'status','data_version','maximum_words'}
print('Public startup record:',public)
print('Missing version:',missing)
print('Absent server-only placeholder:',absent_secret)
print('Changed public length setting:',changed)
print('Mock private values absent from every emitted record:',True)
print('No credential store, signing, rotation, deployment or real environment access is
implemented.')
```

Expected records, missing-value checks and limits

1. Predict the public and blocked-startup records before running or tracing. The script takes supplied mock dictionaries and never reads your computer's environment. It validates required values, refuses a missing placeholder instead of inventing a default, and constructs a separate public record from reviewed non-secret settings.

   It checks that both tested mock private values are absent from every serialized emitted record; it never prints internal state. A successful mock does not establish credential strength, real protected storage, rotation, all possible leak paths or actual answer-length enforcement. Keep those claims outside the observed evidence.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: diagnose a value exposed through a browser error

> Apply the public/private boundary to a different output channel.

Complete independent browser-error scenario

Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

First attempt and support record

> **My independent explanation, trace and limits**
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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

> **My independent procedure, result and limitation**
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the configuration contract and output-field policy

> Keep public settings, required private inputs and redacted diagnostics distinct.

Project task and artifact

Add a labelled **configuration** and **secret**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **configuration** and **secret**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a clear startup requirement and a bounded result

> Report missing keys without displaying privileged values.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **configuration** and **secret**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Secret	A value whose disclosure can grant privileged access
Configuration	Declared settings controlling operation
Redaction	Removing sensitive values from displayed or recorded output

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Configuration and secrets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A startup script prints every environment variable while debugging a missing data version.	The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.	It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

By the end, I can	How I will show it
Explain the mechanism and terminology of configuration and secrets	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.
- Which labelled step matches this detail? Record startup success without printing the token.
- 1. Load credential
- 2. Log safely
- 3. Load setting
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Log safely Record startup success without printing the token. **Configuration** controls behaviour. A credential grants access and needs different storage and handling.

Changed-case question and optional hint

- Changed-case practice: Change maximum_words to 60. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The configured answer limit becomes 60; the private token stays unchanged.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Repair and retain the record

Review Typed contracts and schemas (ai-6/3). Goal: Explain the **mechanism** and terminology of typed contracts and schemas. Short practice: Compare A’s count 4 with B’s count "4".

Which record passes, which gate rejects B, and why does changing A’s count to -1 create a different failure?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor’s request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Configuration and secrets: the case



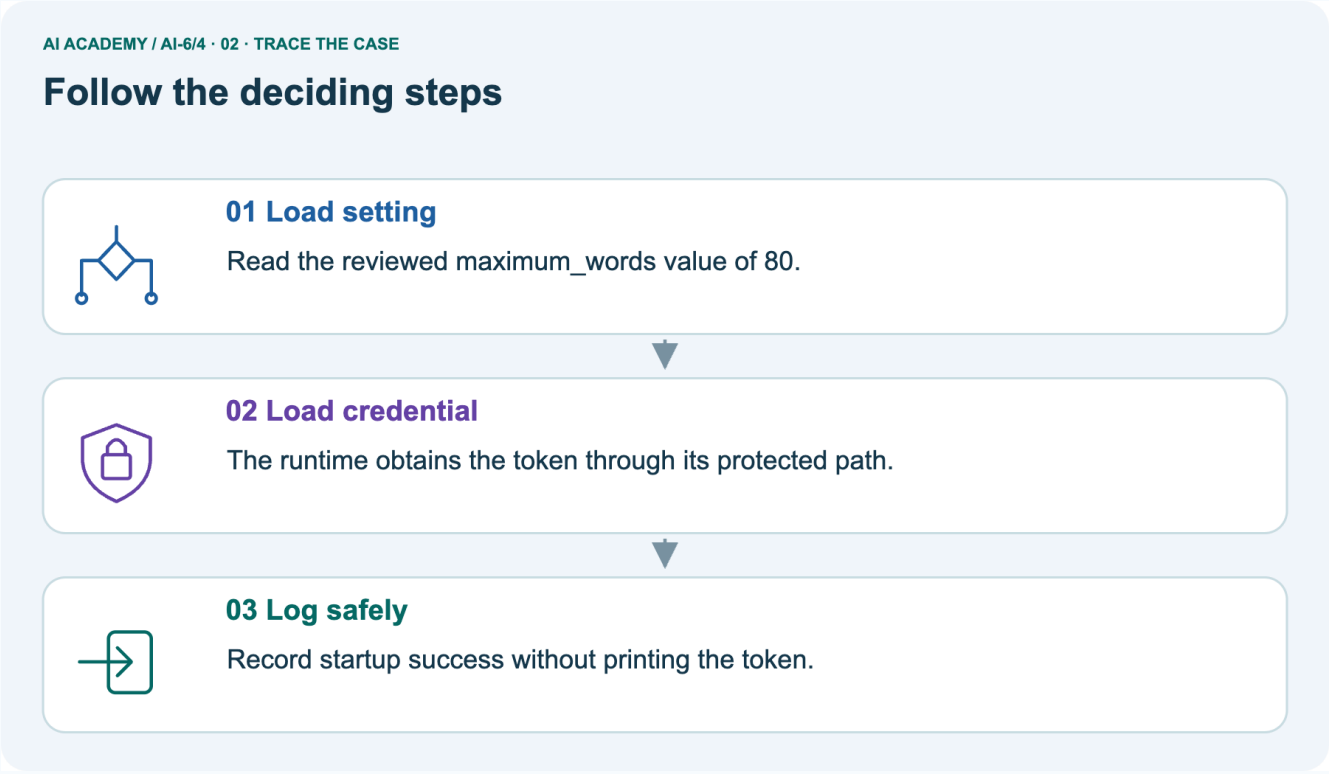
Follow the labelled picture with the matching explanation. Open the vector diagram

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token. Supplied facts: maximum_words=80 is public **configuration**.

A separate protected store holds a placeholder credential.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

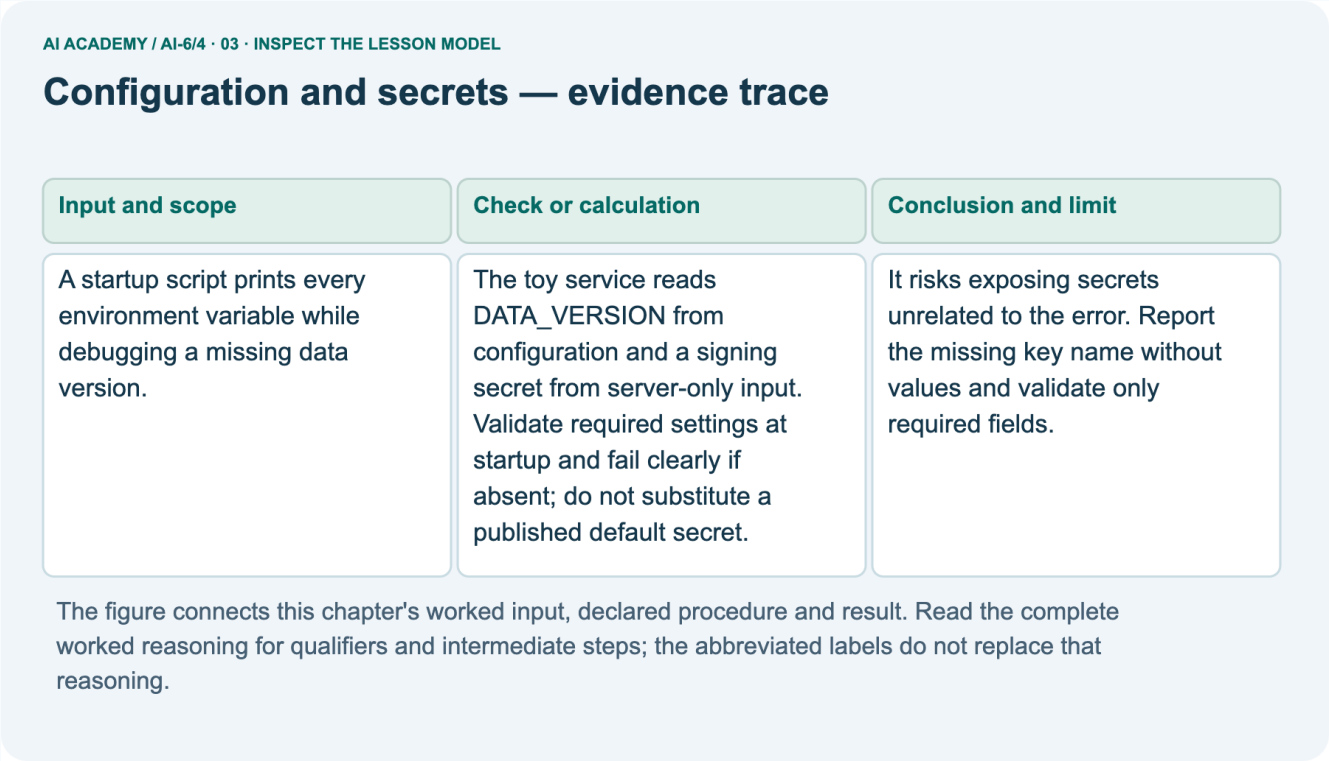


Follow the labelled picture with the matching explanation. Open the vector diagram

Load setting: Read the reviewed maximum_words value of 80. → Load credential: The runtime obtains the token through its protected path. → Log safely: Record startup success without printing the token.

Point to each numbered stage and explain which recorded **input** it uses.

03. Configuration and secrets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration and **secret**s — **evidence** trace.

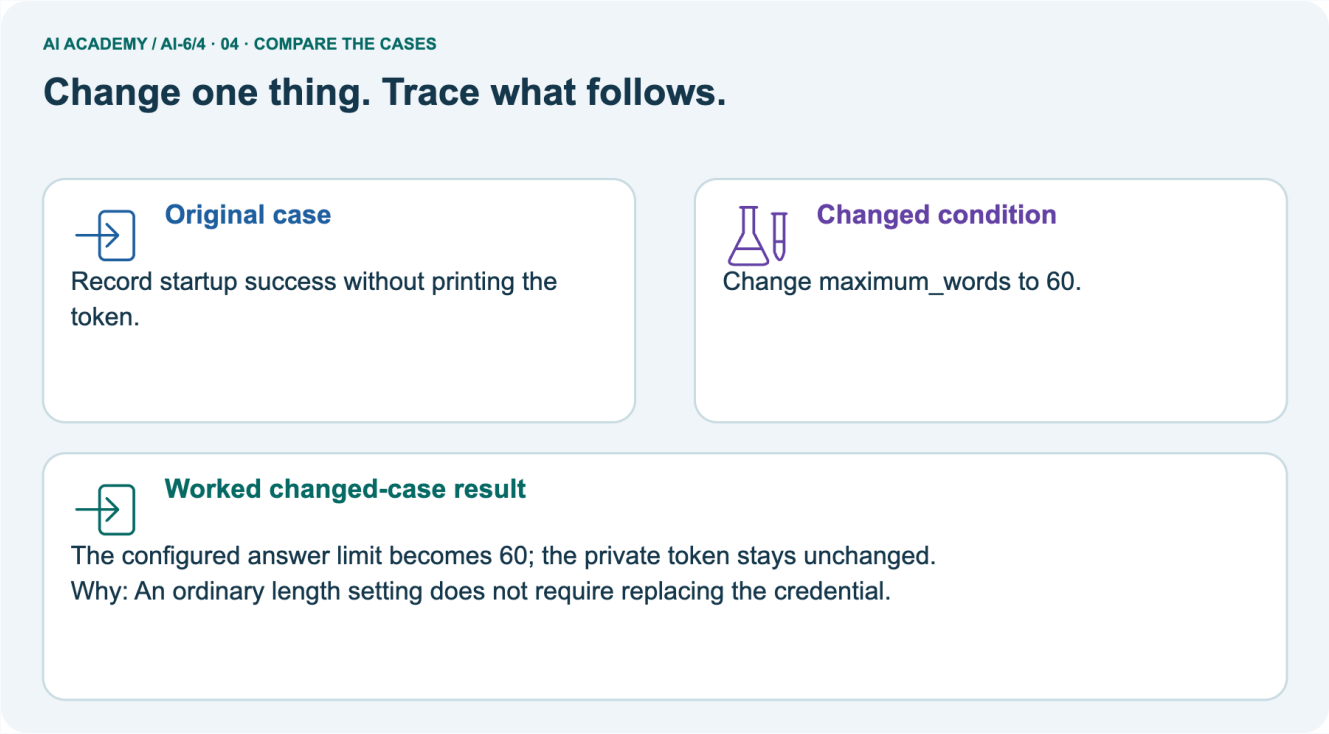
A startup script prints every environment variable while debugging a missing data version. | The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**.

Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**. | It risks exposing **secret**s unrelated to the error.

Report the missing key name without values and validate only required fields.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



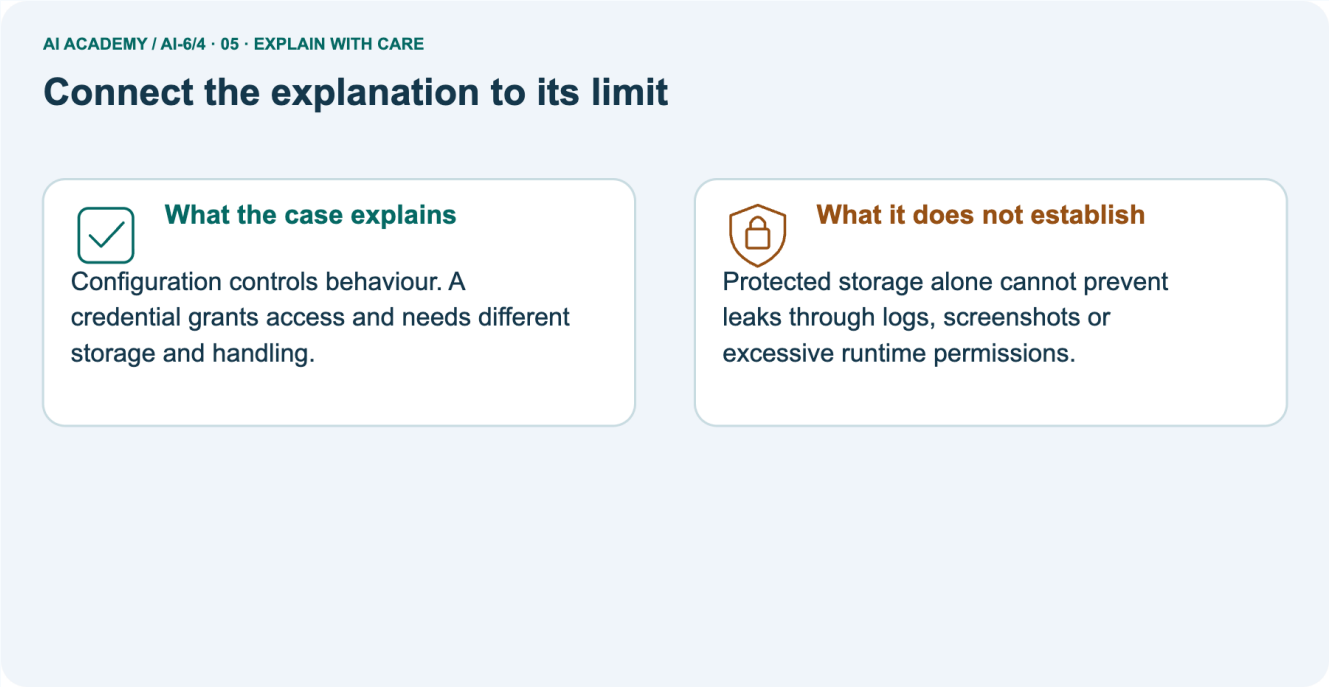
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record startup success without printing the token. Changed condition: Change maximum_words to 60. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

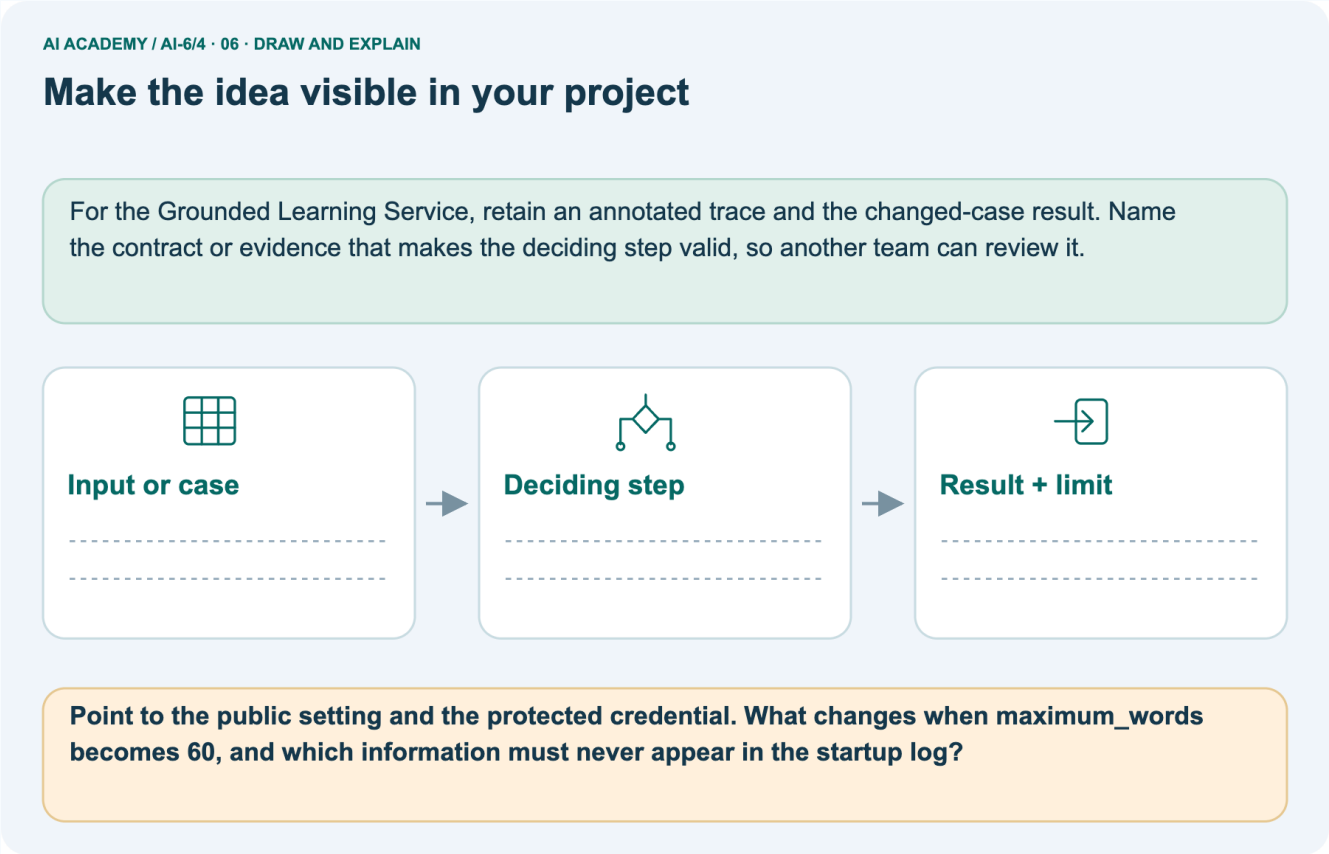


Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration controls behaviour. A credential grants access and needs different storage and handling. Limit: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Configuration and secrets: the case

02. Follow the deciding steps

03. Configuration and secrets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.